

Editorial assessment of “Closed-Loop feedback control system for EXOhSPEC”

RASTI fit and compliance

Desk view: the subject is a strong RASTI fit, but the manuscript is not yet submission-ready. RASTI explicitly covers astronomical instrumentation, sensor systems, instrument development, data methods and associated modelling. A laboratory demonstration of feedback control for a high-resolution spectrograph falls squarely within that scope, particularly because the paper is framed as an instrumentation and validation study rather than as an exoplanet radial-velocity result. ¹

There is no formal RASTI article category called a “Techniques paper”. The relevant category is **Research article**. RASTI publishes Research articles, Invited Reviews, Comments and Corrections; it imposes no page limit on Research articles and explicitly allows shorter technical papers, provided that all other requirements are met and the presentation remains concise. The present 13-page paper, with 13 figures, three tables, a full control-law derivation and several experimental configurations, reads as a full Research article, not as a short technical note. The figure count is not inherently excessive because RASTI sets no formal figure limit; the editorial issue is whether each figure advances the central argument. ²

The manuscript’s central content is appropriate for the journal: it documents a sensing–decision–actuation architecture, reports a same-hardware open-loop comparison, decomposes the actuator contributions and carefully declines to infer radial-velocity precision from centroid motion alone. Those are exactly the kinds of methodological and instrument-validation distinctions RASTI was established to accommodate. ³

Compliance findings

Item	RASTI requirement	Manuscript status	Editorial assessment
Article type and length	Research articles have no page restriction but should be concise.	13 pages, 13 figures and three tables.	Formally acceptable as a full Research article, but over-expanded in the Introduction, Discussion and future-work material.
Abstract	Normally no more than 250 words, one paragraph, summarising aims, methods and new results.	Approximately 333 words .	Non-compliant and likely to be returned for correction before consideration.

Item	RASTI requirement	Manuscript status	Editorial assessment
Structure	Numbered sections; Introduction first; Conclusions last; unnumbered end matter.	Correct broad structure, with Introduction through Conclusions followed by Acknowledgements, Conflict of Interest and Data Availability.	Broadly compliant.
Keywords	At least one journal-supplied keyword, at least two author keywords, six total maximum.	Six keywords are shown.	Likely acceptable, but the ScholarOne metadata must include an exact journal-list keyword such as "Instrumentation".
Conflict of interest	Required in the end matter.	Present: "The authors declare no conflict of interest."	Compliant.
Data Availability	Mandatory; should explain access and, where possible, link to data or software.	Present, but no repository, identifier, code release or assured access mechanism is supplied.	Formally present but editorially weak.
Funding	Funding information or the exact no-funding declaration must appear at the end.	No Funding section or no-funding declaration appears.	Non-compliant.
Figure accessibility	Alt text is required for every figure, placed beneath the legend in the manuscript.	No alt text is visible for any of the 13 figures.	Non-compliant.
British English and punctuation	British spelling; RASTI does not use M-rules/three-dash em rules.	Mostly British spelling, but with several exceptions and extensive dash-parenthetical prose.	Requires a house-style pass.

The abstract limit, end-matter requirements, keyword rules, figure requirements and manuscript structure are stated explicitly in the RASTI author instructions. ⁴ The observations about the submitted manuscript are based on the attached 13-page PDF. [filecite turn0file0](#)

The abstract is the clearest technical-compliance failure. It is about 333 words, approximately one-third above RASTI's normal 250-word maximum. It also tries to carry two papers' worth of results: the initial actuator-decomposition/open-loop study and the later 88.6-hour extended-architecture study. The final sentence alone introduces a new architecture, a second duration, two additional scatter values, an intervention claim, a correction count and an overshoot statistic. That density makes the abstract read as a compressed Results section rather than a selective statement of the paper's main contribution. [filecite turn0file0](#)

A compliant abstract should probably centre on four points only: the control architecture; the 31.44-hour hybrid performance; the matched open-loop degradation and actuator decomposition; and the conclusion that image-position control is distinct from optical-path control. The 88.6-hour result can be reduced to one short confirming clause or omitted if the extended architecture is repositioned as a secondary validation.

The Data Availability statement is unlikely to satisfy an instrumentation editor without further explanation. It says that summary products are “retained in the EXOhSPEC project record”, that the raw telemetry, frames, geometry and calibration files “are not publicly released”, and that material “can be considered” subject to author, laboratory and collaboration approval. That gives readers neither a persistent identifier nor a defined route to obtain the data. RASTI strongly encourages release of the data and software on which conclusions rely, supports FAIR principles, and requires authors withholding data to justify that decision in the submission cover letter. ⁵

The proprietary optical geometry may reasonably remain restricted, but the paper’s conclusions do not require the complete proprietary geometry. A minimal reproducibility package could contain the time stamps, anonymised centroid series, OPL residual, environmental channels, actuator-command ledger, phase labels and scripts needed to regenerate Tables 2–3 and Figs 5–10. Without that package, a referee cannot independently inspect the eleven-fold comparison, the command counts, the periodograms or the 7.9 per cent overshoot classification. The present wording also makes access discretionary rather than reproducible.

The figures need an editorial rather than merely technical reduction. Figures 2–4 establish the hardware and control architecture effectively. Figures 5–10 carry the empirical evidence. Figures 11–13 then introduce three more architecture diagrams after most of the results have already been presented. There is no formal figure-count problem, but the paper currently asks the editor to absorb several overlapping representations of the same controller: a signal-flow diagram, a per-cycle decision flow, a fast-lane flowchart, a timing diagram and a connection-health state machine. At least one could move to supplementary material, while every figure requires alt text before submission. RASTI also asks for vector graphics for plots and colour-blind-friendly design. ⁶

Venue recommendation: RASTI is the best match in the manuscript’s present conceptual form. PASP would also be credible because it explicitly publishes refereed instrumentation articles and positions itself as a technical astronomy journal. ⁷ A&A includes instrumentation and methods within its coverage, but this paper would be more competitive there if it connected the controller to a broader astronomical performance outcome, such as a wavelength solution, extracted spectra or a quantified calibration impact. ⁸ AJ is a weaker fit because, although it accepts instrumentation and analysis techniques, its stated emphasis is on significant astronomical results derived from observations. ⁹

The title also needs attention. The main title is “**Closed-Loop feedback control system for EXOhSPEC**”, whereas the running title is “**Closed-loop thermal-optical control of EXOhSPEC**”. The latter is more precise and more professional. “Closed-loop feedback” is partly redundant, and “Closed-Loop” should not have an internal capital on *Loop* in sentence case. ¹⁰

Jargon and accessibility

The paper has an unusual accessibility burden because it sits between astronomical spectroscopy, optical metrology and control engineering. A RASTI instrumentation referee will accept specialist vocabulary, but the terms must be introduced functionally rather than only named. RASTI’s house style

also requires acronyms and abbreviations to be expanded at first occurrence unless they are genuinely universal across astronomy and geophysics. ¹⁰

Jargon that is appropriate and generally handled well includes *fibre-fed echelle spectrograph*, *image centroid*, *optical-path length*, *refractive index*, *wavelength solution*, *radial-velocity observable*, *PID* after expansion, *ridge regression*, *coefficient of determination* and *periodogram*. The paper often does a good job of placing boundaries around those terms. For example:

“a single feature position is not, by itself, a wavelength solution or a radial-velocity observable.”

That sentence is accessible, scientifically disciplined and useful to readers outside precision spectroscopy. Likewise, the definition of feedback control beginning “A closed-loop, or feedback, control system is one in which...” is clear, although longer than necessary for this venue. [filecite turn0file0](#)

The following terms are more problematic.

“ECU and BME channels” are introduced in Section 2.1 without expansion. The prose tells the reader that they record temperature, pressure and relative humidity, but not what ECU and BME stand for, whether they are sensor packages, internal software labels or manufacturer-specific devices. “IDS3010” and “Meerstetter TEC controller” are also given as product names before the role, measurement precision and sampling relevance are fully established. A cleaner formulation would be: “an IDS3010 displacement interferometer monitors a fixed metrology path; internal and external environmental sensor packages record temperature, pressure and humidity; and a Meerstetter thermoelectric controller adjusts the enclosure set point.” [filecite turn0file0](#)

“Plant” is standard control-engineering language but not universal astronomy language. The sentence

“The plant here combines a slow, nonlinear thermal channel with a fast but discrete actuator...”

will be understood by a controls referee but may momentarily confuse an optical-instrumentation reader. Define it once as “the controlled spectrograph and actuator system (‘the plant’ in control terminology)” or simply avoid the word.

“Gain-scheduled proportional-integral-derivative law” is acceptable, but the manuscript quickly adds *anti-windup*, *feedforward*, *deadband*, *minimum step*, *rate limit*, *sign-flip suppressor* and *load-absorption term*. The individual ideas are not incomprehensible, but the presentation assumes a controls vocabulary that the preceding tutorial on basic open versus closed loop does not prepare the reader for. The sentence

“The commanded increment then passes a fixed cascade before it reaches the hardware...”

is followed by five safeguards in one list. A referee would benefit from a compact controller-parameter table giving each gate, its numerical threshold, its purpose and whether it was exercised in the reported runs. [filecite turn0file0](#)

“Temperature-equivalent error” needs a more physical definition. Equation (8) combines OPL-derived and pixel-derived errors, but the paper does not first tell a non-specialist how each has been converted

into temperature units, what sign convention applies or why a fixed 0.80/0.20 weighting is appropriate. “Favours the direct optical-path estimate” explains the intention, not the calibration.

“Fast-channel load-absorption term” and “ledger” are especially opaque. The manuscript says:

“The cumulative fast-channel state is tracked as a ledger and gently unloaded toward centre...”

A reader can infer the basic idea, but “ledger”, “unloaded” and “released load transferred to the slow channel” are metaphors. State the physical operation: accumulated mirror-stage displacement is periodically reduced, while the thermal set point is adjusted to preserve the detector centroid.

“Bounded relay actuator” is used to characterise the coarse stage. This is defensible control terminology, but a short gloss is needed: the actuator issues discrete signed steps and can alternate around the target because a single step can exceed the residual error.

“Model seed”, “reference tracker”, “quiet”, “armed” and “settle-gate” function as software-language shorthand rather than defined scientific variables. “Quiet” appears in the flowchart condition “Fast channel armed, quiet, and enabled?” but no reader-facing quantitative definition accompanies the word at that point. “Re-freezes” in the connection-health discussion is even less transparent: it appears to mean “does not reset or suspend adaptation of the environmental model”, but that should be stated directly. turn0file0

“Fast AO lane” is potentially misleading. In astronomy, AO overwhelmingly suggests *adaptive optics*. Here it appears to mean *active optics*, or simply the fine-optics actuator. The manuscript does not consistently expand the acronym in the running text. Since this is neither atmospheric adaptive optics nor a wavefront-correction system in the conventional sense, I would avoid “AO” altogether and use “fast fine-optics channel”. This would also align the terminology across the abstract, Sections 3–5 and the figure labels.

The distinctions among “image position”, “image displacement”, “centroid”, “radial error”, “scatter”, “stability”, “RMS” and “RMS from the origin” need stronger policing. Section 2.2 correctly distinguishes scatter about the local mean from RMS distance from the fixed reference. Later, however, prose such as “degrades stability to 0.316 pixels” does not say which statistic or coordinate is meant. In Tables 2–3, the principal figures are $\sigma(\Delta Y)$ and $\sigma(\Delta X)$; in other places the paper refers to radial RMS. The abstract should say “vertical centroid scatter” rather than the generic “stability”.

turn0file0

The paper is clearest when it uses concrete laboratory language. For example:

“The fast channel is held off for a fixed interval after any thermal action...”

and

“Frames with pixel outliers are excluded from the regression...”

are immediately understandable. It becomes less accessible when several layers of interpretation are compressed into one sentence, as in:

“This cascade converts the analytically optimal effort into a physically admissible, non-oscillatory setpoint change appropriate to a slow thermal actuator.”

“Analytically optimal” is not demonstrated in the preceding text; “physically admissible” and “non-oscillatory” each conceal multiple implementation constraints. A plainer sentence would be more credible: “These gates limit the calculated correction to steps that the thermal hardware can apply without immediate reversal or excessive set-point changes.”

The Introduction is also unevenly pitched. It spends a full paragraph explaining the elementary distinction between open- and closed-loop control, yet later assumes familiarity with ridge regularisation, gain scheduling, anti-windup and hysteresis without equivalent explanation. The better balance is to shorten the basic feedback tutorial and use the saved space to explain the manuscript-specific controller terminology.

Does it read as AI-generated?

No textual assessment can establish whether a manuscript was written by a person, generated by a model or revised with language tools. The relevant editorial question is whether the prose exhibits patterns that make it feel machine-polished, formulaic or insufficiently owned by the authors.

My assessment is that it does not read as fabricated science, but substantial parts read as heavily machine-edited or assembled from revision-response prose. The laboratory-specific detail, numerical caveats and explicit limits on interpretation feel grounded in a genuine research programme. The AI-like impression comes mainly from repeated meta-summary, contrast formulas and reader-management language.

The strongest tell is the presence of **revision correspondence inside the article**. The Introduction says:

“the central addition of this revision”

The Conclusions begin:

“New in this revision, we also decompose that result...”

and later:

“New in this revision, an extended architecture...”

The Discussion says that the open-loop comparison provides evidence “in a way the earlier draft of this work could only assert”. These are not journal-article sentences; they are sentences from a response to reviewers or an internal change log. Their presence makes the paper look incompletely converted from a revision exercise into a standalone publication. fileciteturn0file0

A second tell is the repeated use of **formulaic paragraph openers that tell the reader what to think before presenting the evidence**:

“Read together, these two results indicate...”

"This result is nonetheless useful..."

"This is the central claim of this section..."

"Sections 3 and 4.1 together show two things..."

"Of the two new comparisons, the open-loop result matters more."

"A separate, exploratory measurement supports the same actuator picture from a different angle."

"The same validation approach used for the thermal control law extends to actuator calibration itself..."

"Together these convert the present bounded-feedback and actuator-decomposition demonstration..."

Any one of these would be normal. Their accumulation creates an LLM-like "summarise, interpret, re-summarise" rhythm. The prose repeatedly announces the significance of a result rather than allowing the result, uncertainty and comparison to establish that significance. turn0file0

A third tell is the high frequency of **binary rhetorical constructions**:

"rather than serving as performance benchmarks..."

"not incidental telemetry but a prerequisite..."

"not by design intent alone..."

"a correlation, not an established causal link..."

"not recovery of a stationary optical path..."

"not only over the original 31-hour one..."

This "not X, but Y" structure is useful for limiting claims, and several instances are scientifically valuable. Repetition makes the paper sound optimised for defensiveness. Vary the syntax and retain only the distinctions that materially prevent misinterpretation.

A fourth tell is the tendency towards **long, balanced sentences with multiple qualifications and an explanatory tail**. For example:

"The most plausible reading is that the controller was responding to a genuinely evolving disturbance with a model that was no longer valid, not that a single corrective action itself created the drift..."

The sentence is careful, but it combines interpretation, exclusion of an alternative and retrospective justification. Similar structures recur throughout the Discussion. Breaking them into shorter evidence-led units would create a more natural research voice.

The paper also overuses dash-parenthetical insertions. RASTI permits N-rules as parentheses but explicitly does not use M-rules. Beyond the house-style issue, the repeated pattern “claim – qualification – further claim” contributes to the machine-polished cadence. ¹¹

There are, however, several passages that sound distinctly like a genuine researcher describing the limits of a real apparatus:

“a small scatter in these quantities does not prove that the origin itself was recovered.”

“Both single-actuator runs are short relative to the hybrid record... and their statistics should be read as indicative rather than definitive.”

“This periodicity is reported as a correlation, not an established causal link...”

“The connection-health state machine... was not exercised by a real channel fault during this run.”

“we do not infer a radial-velocity precision from these measurements.”

These sentences show scientific restraint, awareness of measurement boundaries and knowledge of what the experiment did not establish. The exact command counts, phase durations, thresholds and failure modes also give the paper a real laboratory voice. [filecite00turn0file00](#)

The best editorial remedy is therefore not to “de-AI” the paper through cosmetic synonym changes. It is to remove revision-history wording, cut the repeated interpretive signposts, shorten the Discussion and make the evidence sequence do more of the argumentative work. After that, the prose would read as an instrumentation paper written by researchers rather than a manuscript repeatedly explaining its own rhetorical purpose.

Narrative coherence — can a reader follow the project start to finish?

The core project is followable, but the paper currently contains two overlapping narratives. The first is clean and persuasive: establish the testbed, define the closed loop, compare hybrid operation with individual actuators and with open loop, then discuss what is and is not controlled. The second is a later V17 or “extended architecture” story involving a finer actuator, fast-lane hysteresis, connection-health monitoring, an 88.6-hour run and overshoot statistics. Those stories have not yet been fully integrated.

The Introduction identifies the right scientific-editorial question, but reaches it too slowly. The first paragraph correctly positions complete-system spectrograph stability and limits the paper to centroid displacement and one interferometric path. The second paragraph gives relevant comparator instruments. The refractivity paragraph is useful physical motivation. After that, however, the manuscript pauses to teach generic feedback-control concepts, lists the architecture, then provides a second control-engineering framing before finally asking the main question:

“can a time-synchronised feedback loop... hold the measured image position within a controlled region, and how much of that result comes from each actuator individually...?”

That is the correct organising question. It should appear near the end of the second or third paragraph, not after an extended tutorial. turn0file0

The Introduction also makes a causal claim too early in the Figure 1 caption. It says that smoothly evolving pressure and humidity are “isolating temperature as the driver of this particular excursion”. A time-coincident temperature dip is evidence consistent with a thermal contribution, but “isolating” sounds stronger than the observational design described. Editorially, “the temperature dip is the only recorded environmental feature coincident with the OPL excursion” is safer and more precise.

Sections 2 and 3 mostly build logically. Section 2 defines the observables, origin and metrics; Section 3 provides the environmental models, fused control law, fine-actuator mapping, calibration and gates. This is the strongest structural portion of the paper. Equation (4)’s distinction between scatter about a mean and RMS from the fixed origin is particularly important.

There are nevertheless two method-level discontinuities. First, the paper says that the matrix in equation (10) corresponds to approximately 0.16 px per step and commands at most one step per cycle. Section 3.4 then introduces a replacement actuator roughly 30–35 times finer, used in the 88.6-hour run, for which corrections involve tens of steps. The reader is therefore dealing with two materially different fast-actuator regimes under a single “fast channel” label. A version table should state, for every run, the hardware revision, matrix, step size, thresholds, cadence and software architecture.

Second, the calibration acceptance rules are internally unclear. Section 3.4 first specifies:

“a working acceptance threshold of $\theta_{\text{err}} < 5^\circ$ per axis.”

It later reports all four directions as being within:

“a 5–15° per-axis acceptance band”

with two passing and two receiving marginal warnings. A hard threshold below 5° and an acceptance band extending to 15° are not the same criterion. Define separate **pass**, **warning/operationally acceptable** and **fail** bands, and use them consistently in the methods, figures and Discussion.

turn0file0

Section 4.1 is the manuscript’s clearest answer to the question posed. It presents the configurations together, gives a common table and distinguishes the short single-actuator tests from the longer hybrid record. The same-session open-loop phase is an editorially strong contribution because it avoids relying solely on a historical comparison.

However, the opening sentence is inaccurate:

“Four closed-loop configurations... were run...”

The list includes an open-loop interval, and the table contains five rows. A more accurate description is “four active configurations and one open-loop interval”, or “five operating phases, including three principal actuator configurations and a matched hybrid/open-loop sequence”.

More importantly, the command-count interpretation contradicts Table 2. The paper says the fast-only configuration issued:

“an order of magnitude fewer actions per unit time than the thermal channel manages on its own.”

Table 2 gives 13 fast-only commands over 4.06 h, approximately **3.2 commands h⁻¹**, and five thermal-only commands over 2.95 h, approximately **1.7 commands h⁻¹**. The fast-only command rate is therefore roughly 1.9 times the thermal-only rate, not one-tenth of it. The Discussion repeats the problem by describing the fast-only configuration as operating at “one-tenth the command rate of the thermal-only configuration”. [filecite turn0file0](#)

The abstract’s “order of magnitude fewer corrective actions” appears to compare raw totals across runs of radically unequal duration—13 commands in 4.06 h versus a much larger total in 31.44 h. That comparison is not meaningful without normalisation. This is not a minor copy-edit: it is a prominent interpretive claim repeated in the Abstract, Results, Discussion and Conclusions while being inconsistent with the table. It must be audited before peer review.

The open-loop comparison also needs a cadence explanation. The 23.31-hour open-loop interval contains only 106 frames, whereas the 32.24-hour preceding hybrid phase contains 1996 frames and the nominal cadence elsewhere is approximately 70 s. A reader will ask whether open-loop frames were intentionally sampled much more sparsely, whether exposures were rejected or whether the phase duration includes gaps. Because time-dependent scatter is the central comparison, the sampling difference should be disclosed and its implications addressed. [filecite turn0file0](#)

Sections 4.2 and 4.3 are coherent secondary analyses but partially divert the narrative. The evolving OPL result directly supports the paper’s important distinction between detector-plane control and path-length control. The outdoor-temperature predictor test is a reasonable negative result. Yet the manuscript gives this predictor a full subsection while the much more consequential sampling, actuator-version and command-rate questions remain underexplained. The outdoor-temperature subsection could be shortened and folded into the model-validation discussion.

The sentence

“It also suggests, tentatively, that residual instability... owes more to actuator saturation...”

is appropriately hedged but not strongly supported by the result presented. The failure of one additional predictor to improve R^2 does not by itself establish actuator saturation as the leading explanation. From an editorial perspective, either provide the saturation evidence or stop after the negative model result.

Section 4.4 breaks the methods-before-results sequence. It reports the 88.6-hour performance of an “extended architecture described in Section 5.1”. That architecture—hysteresis, fast cadence, unload logic and connection-health states—is not actually explained until the Discussion, after the results. Figure 13 is referenced in the Results before its state machine is presented. A first-time reader is asked to interpret 699 corrections, threshold exceedances, settle intervals and health-state behaviour without first having the full algorithm. [filecite turn0file0](#)

Section 5.1 and Figs 11–13 should move into the Control Algorithm section, probably as “Extended V17 controller”. Section 4.4 would then report its validation. The Discussion should interpret rather than introduce architecture.

The phase accounting in Section 4.4 is also hard to reconstruct. It calls the run 88.6 h of closed-loop operation, says the full logged record including 2.61 h of warm-up spans 93.1 h, identifies 4861 feedback frames, says 4761 are settled and reports 141 warm-up frames in Table 3. The missing transitional duration and frame population can be inferred, but it should not have to be inferred. A single phase table should list warm-up, transition, settled feedback and any excluded intervals, with start/end times, durations, frame counts and exclusion rules.

The health-state result is overstated. The paper first correctly says that no real channel fault occurred. It then says:

“The degrade/recover path... remains validated by code inspection and by the absence of any unhandled fault...”

The absence of a fault cannot validate recovery from a fault. The proper conclusion is: the state machine was implemented and inspected, but its degrade/recover transitions were not empirically exercised. Calling it “validated” and then immediately saying it was not tested creates avoidable distrust.

The Discussion contains evidence that should either have appeared earlier or should be removed.

Its opening paragraph introduces an earlier failed run with a full-record radial RMS of 5.892 px. That run is not defined in the experimental configurations, tabulated or illustrated. It reads as historical motivation inserted after the results. Either document it in a short background subsection with enough information to evaluate it, or omit the numerical anecdote.

Likewise, the “localised thermal-impulse campaign” first appears in the Discussion. New experimental results should not debut there. If the result matters to the actuator interpretation, give it a concise Results subsection and show the evidence. If it is only preliminary context, reduce it to one sentence under future work.

Section 6 makes the manuscript feel less complete than it is. A full numbered section titled “Planned validation experiments” reads like a project proposal. Some future work is appropriate, but the present section lists repeats, restart tests, longer single-actuator runs, time-constant measurements, open-loop OPL monitoring, fault injection and another orthogonality verification. This catalogue reinforces the impression that the manuscript is an interim system report.

Retitle the section “Limitations and further validation”, move it into the Discussion and reduce it to the two or three limitations that directly qualify the current claims: unequal single-actuator durations, unexercised fault recovery and absence of a wavelength/RV validation. The detailed development roadmap belongs in internal documentation.

The Conclusions match the broad evidence but are too long and repetitive. They reproduce most of the Abstract and Results, include “New in this revision” twice and then append an extended V17 mini-summary. A stronger conclusion would contain four concise findings: the matched open-loop benefit; the actuator decomposition, with corrected command-rate wording; the distinct OPL behaviour; and the 88.6-hour confirmation. It should end with the scope boundary, not a full future-work list.

Grammar and language

The manuscript is predominantly in British English—*stabilised, fibre, behaviour, centred* and *programme* are generally used correctly. RASTI requires British spelling, uses “per cent” in prose, specifies standard

unit abbreviations such as h and μm , and does not use M-rules. ¹² The following are concrete corrections from the submitted PDF. turn0file0

Location and original wording	Problem	Suggested revision
Title: “Closed-Loop feedback control system for EXOhSPEC”	Inconsistent capitalisation; “closed-loop feedback” is redundant; differs from running title.	“Closed-loop thermal-optical control of EXOhSPEC”
Abstract: “Disabling each actuator in turn isolates their individual roles”	Singular <i>each</i> does not agree cleanly with plural <i>their roles</i> .	“Disabling the actuators in turn isolates their respective roles”
Abstract: “degrades stability to 0.316 pixels”	Metric and coordinate are undefined.	“increases the vertical centroid scatter to 0.3164 px”
Abstract: “matches the full hybrid result, 0.138 pixels”	Awkward apposition and stronger than the numerical comparison warrants.	“gives a comparable vertical scatter of 0.1378 px”
Abstract: “using an order of magnitude fewer corrective actions”	Misleading comparison of unequal-duration run totals; contradicted when rates are calculated.	Remove unless a valid normalised comparison is supplied.
Introduction: “benchmarks an ambient-air testbed should match”	Missing relative conjunction; compressed phrasing.	“benchmarks that an ambient-air testbed should be expected to match”
Introduction: “has local room-condition sensitivities”	Unidiomatic construction.	“has sensitivities under local room conditions”
Introduction: “by of order a micron”	Ungrammatical combination of <i>by</i> and <i>of order</i> .	“by approximately 1 μm ” or “by an amount of order 1 μm ”
Figure 1 caption: “localized excursion”	American spelling.	“localised excursion”
Figure 1 caption: “isolating temperature as the driver”	Causally over-strong wording.	“making the temperature dip the only recorded coincident environmental feature”
Throughout: “thermo-electric” and “thermoelectric” concepts	Hyphenation/word choice varies.	Use “thermoelectric” consistently unless naming a specific device.
References and citations: “Å ström”	Erroneous space within the surname.	“Åström”

Location and original wording	Problem	Suggested revision
Section 3.1: “design matrix X carrying the five predictors”	The base model has four environmental predictors before the later outdoor-temperature term; model versions are conflated.	Specify “four predictors in the baseline model and five in the extended model”.
Section 3.1: “as temperature and refractivity necessarily are”	Refractivity is not clearly one of the stated predictor columns.	Name the actual collinear predictors or delete the clause.
Section 3.2: “analytically optimal effort”	“Optimal” has not been defined or demonstrated.	“calculated controller output”
Section 3.5: “The fast channel is held off for a fixed interval after any thermal action, relaxed after a small step, because...”	Dangling modifier: the channel is not “relaxed”; the interval or rule is.	“The hold-off interval is shortened after a small thermal step because...”
Section 3.5: “a correction... would be applied against a disturbance about to reverse”	Predictive assertion presented as universal.	“may be mistimed if the transient reverses before the plant responds”
Section 4.1: “Four closed-loop configurations”	The list includes an open-loop interval and the table has five rows.	“Five operating phases, including four active phases and one open-loop interval...”
Section 4.1: “an order of magnitude fewer actions per unit time”	Numerically inconsistent with Table 2.	Recalculate and rewrite from a defined denominator.
Section 4.2: “The dependence between a controlled image position and a still-evolving optical path...”	<i>Dependence</i> implies a demonstrated relationship; the text describes coexistence.	“The coexistence of a bounded image position and an evolving optical path...”
Section 4.3: “This result is nonetheless useful”	Formulaic meta-commentary.	State the direct conclusion: “The additional predictor did not improve the model in this run.”
Section 4.4: “the longest continuous record in this programme by a wide margin over...”	Wordy comparative construction.	“the longest continuous record in the programme, compared with the previous 31.44-h run”
Section 4.4: “a weaker harmonic near 17.7 h ($\approx 2\times$)”	17.7/8.1 is about 2.19, and “harmonic” may be imprecise for the longer period.	“a weaker feature near 17.7 h” unless harmonic status is quantitatively established.

Location and original wording	Problem	Suggested revision
Section 4.4: “plausibly a building HVAC or enclosure duty cycle”	Reasonable hypothesis, but too prominent without supporting environmental evidence.	“potentially associated with a building or enclosure thermal cycle”
Section 4.4: “remains validated by code inspection...”	Code inspection and fault-free operation do not validate unexercised recovery transitions.	“was implemented and inspected but not empirically exercised”
Section 5.1: “two or more channels failing simultaneously degrades further”	Awkward subject-verb structure.	“simultaneous failure of two or more channels causes a further transition...”
Section 5.1: “never re-freezes or resets the environmental model”	“Re-freezes” is unexplained software shorthand.	“does not suspend adaptation or reset the environmental model”
Section 6: “Together these convert...”	The experiments are future plans and cannot yet convert the present demonstration.	“These experiments would extend...”
Conclusions: “New in this revision...”	Response-to-reviewer language, inappropriate in a clean article.	Delete and state the result directly.
Conclusions: “what a given actuator is contributing”	Conversational.	“the contribution of each actuator”

A separate consistency pass is needed for **fast optics**, **fine optics**, **fast-optics stage**, **fine-optics stage**, **fast channel** and **AO lane**. These appear to refer to overlapping or successive hardware/software concepts, but the naming is not stable. Use one term for the actuator and one for the software path, then identify hardware revisions explicitly.

Similarly, use the statistical notation consistently. In prose, “scatter” should map to σ ; “RMS from the operating reference” should map to RMS_0 ; and “radial error” should refer to r . At present, “RMS”, “scatter” and “stability” sometimes substitute for one another even though Section 2.2 correctly defines them as different quantities.

The captions also need shortening. Figure 1’s caption not only identifies four panels but argues for a causal interpretation and motivates the whole controller. Figure 4’s caption narrates almost the entire decision flow. Figure 10 interprets possible HVAC causation. Captions should remain sufficiently self-contained, but interpretive discussion belongs mainly in the body.

Finally, the title and major headings should be checked against RASTI’s sentence-case convention. The supplied class correctly renders major numbered headings in capitals, but the article title itself should not use attention-style internal capitals. RASTI also directs authors to use N-rules rather than M-rules; any three-dash constructions in the LaTeX source should be replaced. ¹³

Overall editorial verdict

Verdict: needs major revision before review.

I would not reject this as the wrong venue. The subject is highly appropriate for RASTI, the same-session open-loop comparison is editorially valuable, and the manuscript displays commendable restraint in refusing to equate detector-centroid stability with a complete radial-velocity result. The paper contains enough substantive instrumentation work to justify peer review after revision.

filecite turn0file0

I would not send the present version directly to referees, however. Some defects are straightforward technical returns: the abstract exceeds the journal's normal maximum by more than 80 words, the required funding declaration is absent, and figure alt text is missing. RASTI states that manuscripts lacking required contents may be returned for correction before publication consideration. ¹⁴

The **single biggest reason for editorial hesitation is that the manuscript does not yet look internally reconciled after successive hardware and controller revisions**. The clearest symptom is the prominent assertion that the fast-only configuration used “an order of magnitude fewer actions per unit time” or “one-tenth the command rate” of thermal-only operation, while Table 2 gives approximately 3.2 fast-only commands h^{-1} and 1.7 thermal-only commands h^{-1} . That contradiction appears in the Abstract, Results, Discussion and Conclusions. At the same time, the calibration section gives incompatible-looking 5° and $5\text{--}15^\circ$ acceptance criteria, the extended controller is described after its results, and an unexercised fault-recovery mechanism is described as validated.

filecite turn0file0

An editor encountering those inconsistencies will worry not only about the individual sentences but about whether all run labels, controller versions, thresholds, frame counts and derived claims have been audited against the underlying logs. That concern is amplified by the restrictive Data Availability statement.

Before resubmission, the manuscript needs five substantive editorial actions:

- Rebuild the article around a single version-and-run table that identifies the actuator hardware, step matrix, controller architecture, thresholds, duration, cadence, retained frames and enabled channels for every experiment.
- Audit every quantitative claim against Tables 2–3 and the logs, especially command rates, phase accounting, open-loop cadence, calibration tolerances and overshoot denominators.
- Move the extended-architecture description and Figs 11–13 before the 88.6-hour Results subsection; convert the current “Planned validation experiments” section into a short limitations paragraph.
- Reduce the abstract to no more than 250 words, remove all “new in this revision” language, shorten the Conclusions and eliminate repetitive meta-summary.
- Supply the required Funding and alt-text material and strengthen the Data Availability statement with at least a de-identified numerical dataset and analysis scripts sufficient to reproduce the central comparisons.

After those changes, I would regard the paper as suitable to send to referees at RASTI. In its current form, it reads less like a wrong-venue submission than like a promising instrumentation manuscript submitted one editorial cycle too early.

2 General Instructions | RAS Techniques and Instruments | Oxford Academic

https://academic.oup.com/rasti/pages/general-instructions?utm_source=chatgpt.com

3 RAS launches new multi-disciplinary journal | The Royal Astronomical Society

https://ras.ac.uk/news-and-press/news/ras-launches-new-multi-disciplinary-journal?utm_source=chatgpt.com

4 6 14 General Instructions | RAS Techniques and Instruments | Oxford Academic

https://academic.oup.com/RASTI/pages/general-instructions?utm_source=chatgpt.com

5 10 11 12 13 General Instructions | RAS Techniques and Instruments | Oxford Academic

<https://academic.oup.com/RASTI/pages/general-instructions>

7 Publications of the Astronomical Society of the Pacific - IOPscience - Publishing Support

https://publishingsupport.iopscience.iop.org/journals/publications-of-the-astronomical-society-of-the-pacific/?utm_source=chatgpt.com

8 Astronomy & Astrophysics – DOAJ

https://doaj.org/toc/0004-6361?utm_source=chatgpt.com

9 Astronomical Journal - AAS Journals

https://journals.aas.org/astronomical-journal/?utm_source=chatgpt.com